LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 6

Course Code: INT396

Unit: 6

Course Title: UNSUPERVISED LEARNING

Unit Title: Evaluation and Applications of Unsupervised Learning

Course Outcome: CO6

Unit Mapped RBT: L3 (Apply)

Paper: Sample Multiple Choice Questions - ANSWER KEY

Total Questions: 40

Maximum Marks: 40 x 1 = 40

Suggested Duration: 60 Minutes

Answer Grid (Key)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1	A	Q11	D	Q21	A	Q31	C
Q2	B	Q12	B	Q22	C	Q32	A
Q3	C	Q13	D	Q23	B	Q33	A
Q4	D	Q14	C	Q24	D	Q34	D
Q5	B	Q15	B	Q25	C	Q35	B
Q6	D	Q16	A	Q26	D	Q36	C
Q7	A	Q17	B	Q27	A	Q37	C
Q8	C	Q18	A	Q28	B	Q38	B
Q9	C	Q19	D	Q29	D	Q39	D
Q10	A	Q20	C	Q30	B	Q40	A

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Name the external validation measure that corrects the Rand Index for the agreement expected by chance.

A) The Adjusted Rand Index <-- correct
B) The silhouette coefficient
C) The Davies-Bouldin Index
D) The inertia of the partition

Answer: A. The Adjusted Rand Index subtracts the agreement two random partitions would show, so a value near zero means no genuine agreement.

Q2. Identify the two properties that a good clustering has to combine.

A) Accuracy and precision
B) Cohesion and separation <-- correct
C) Recall and specificity
D) Support and confidence

Answer: B. Records inside a cluster should be close together and clusters should be far apart. Every internal validity index is some balance of these two.

Q3. State the silhouette value conventionally assigned to a record that is the only member of its cluster.

A) 1
B) -1
C) 0 <-- correct
D) It is undefined

Answer: C. There is no within-cluster distance to compute for a lone record, so the convention is to record a silhouette of zero rather than leave it missing.

Q4. Name the internal index built from the ratio of between-cluster dispersion to within-cluster dispersion.

A) The Adjusted Rand Index
B) The Jaccard index
C) The confidence of a rule
D) The Calinski-Harabasz index <-- correct

Answer: D. Also called the variance ratio criterion, it rewards partitions whose clusters are tight and widely separated; a higher value is better.

Q5. Identify what a cluster profile reports.

A) The distance between every pair of records
B) The typical feature values of each cluster <-- correct
C) The number of iterations the run took
D) The random seed used for the run

Answer: B. Profiling compares each cluster's feature averages with the overall averages, and it is what turns an anonymous label into a describable segment.

Q6. State the value the Adjusted Rand Index takes when two partitions agree completely.

A) 0
B) -1
C) The number of clusters

D) 1 <-- correct

Answer: D. The index is scaled so that perfect agreement gives 1 and agreement no better than chance gives approximately 0.

Q7. Explain why an external validation measure cannot be applied to a wholly unlabelled dataset.

A) It compares the clusters against reference labels <-- correct

B) It requires the features to be standardised first

C) It can only be computed for exactly two clusters

D) It needs the algorithm to be run several times

Answer: A. External measures score agreement with a known grouping. With no such grouping available, only internal measures and stability checks remain.

Q8. Distinguish cohesion from separation in a clustering.

A) Separation is closeness within, cohesion is distance between

B) Both of the two describe closeness within a cluster

C) Cohesion is closeness within, separation is distance between <-- correct

D) Both of the two describe distance between clusters

Answer: C. A partition can be strong on one and weak on the other, which is why a measure combining them, such as the silhouette, is more informative than inertia alone.

Q9. Describe what consensus clustering sets out to do.

A) Run one algorithm until the inertia is zero

B) Compare a clustering against known class labels

C) Combine many runs into one agreed partition <-- correct

D) Reduce the features before the clustering starts

Answer: C. Repeated runs on resampled data are pooled, and records that keep landing together are grouped. Structure that survives this is far more likely to be genuine.

Q10. Interpret an Adjusted Rand Index close to zero between a clustering and a set of reference labels.

A) The agreement is no better than chance <-- correct

B) The two partitions agree almost perfectly

C) The clustering has exactly zero clusters

D) The reference labels were recorded wrongly

Answer: A. Zero is the value expected from two unrelated partitions, so the clustering has found nothing corresponding to the reference grouping.

Q11. Predict what happens to a customer segmentation over a year as buying behaviour shifts.

A) It becomes more accurate as data accumulates

B) It remains valid because it uses no labels

C) It automatically updates itself from new records

D) It drifts out of date and has to be rebuilt <-- correct

Answer: D. The segments describe the behaviour present when the model was fitted. As that behaviour changes, the boundaries stop matching the customers.

Q12. Recognize the ethical risk described: a segmentation avoids using ethnicity but relies on postcode, browsing language and preferred store.

A) The features are too few for a reliable result

B) Proxy features can reintroduce a protected attribute <-- correct

C) The features are all measured on the same scale

D) The features make the clustering run too slowly

Answer: B. Removing the sensitive column is not enough when other retained columns correlate strongly with it. Outcomes must be audited by group before deployment.

Q13. Explain why the sizes of the clusters should always be reported with a clustering result.

A) They are needed to compute the silhouette score

B) They determine the random seed that was used

C) They fix the number of features to retain

D) They expose a degenerate partition that scores well <-- correct

Answer: D. A partition holding 990 records in one cluster and five each in two others can still show a respectable index. The sizes make that immediately visible.

Q14. Describe why the silhouette score should be reported together with the number of clusters chosen.

A) It states how many iterations the run required

B) It states which features were used in the run

C) It states how well supported that particular choice is <-- correct

D) It states the proportion of records left as noise

Answer: C. The number alone tells a reader nothing about whether the groups are well separated. The score is the evidence that the choice was reasonable.

Q15. Explain why an unsupervised model still needs monitoring once it is deployed.

A) The algorithm gradually loses its parameters

B) The data it describes keeps changing over time <-- correct

C) The features become correlated after deployment

D) The absence of labels makes it unstable to run

Answer: B. A segmentation is a description of a population at one moment. Without monitoring, decisions continue to be made on groups that no longer exist.

Q16. Distinguish purity from the Adjusted Rand Index as external measures.

A) Purity ignores chance agreement and rewards many clusters <-- correct

B) Purity corrects for chance while the Rand Index does not

C) Both of the two correct for the agreement due to chance

D) Neither of the two requires any reference labels

Answer: A. Purity rises simply by splitting into more clusters, so it can be driven to one. The adjusted index removes both that effect and the chance baseline.

Q17. Interpret a clustering that has placed 98 per cent of the records into a single cluster.

A) The data holds one extremely strong group

B) The result is degenerate and must be reconsidered <-- correct

C) The silhouette score must therefore be zero

D) The algorithm converged unusually quickly

Answer: B. Such a split says almost nothing about the population. The usual causes are an unsuitable k, unscaled features or one dominant column.

Q18. Predict the effect on interpretability of clustering on principal components rather than on the original features.

A) The segments become much harder to describe <-- correct

B) The segments become much easier to describe

C) Interpretability is unaffected by the change

D) The segments can no longer be given labels

Answer: A. A component is a weighted mixture of many columns, so a cluster centre expressed in components cannot be read back in business terms without extra profiling.

Q19. Explain why transparency matters when customers are segmented on their behaviour.

A) Segmentation is forbidden without written consent

B) Transparency improves the silhouette of the result

C) Transparency reduces the cost of running the model

D) Decisions are being made about people who cannot see them <-- correct

Answer: D. A person offered a different price or product because of a hidden group assignment has no way to question it. Disclosure and a route of appeal are the safeguards.

Q20. Recognize the case study described: images of handwritten digits are projected into two dimensions to see whether the ten digits separate visually.

A) An association rule mining of purchase baskets

B) An anomaly detection study of network traffic

C) A dimensionality-reduction visualisation of MNIST <-- correct

D) A hierarchical clustering of customer records

Answer: C. MNIST is the standard demonstration for PCA, t-SNE and UMAP, because the true digit labels allow the picture to be checked afterwards.

Q21. Apply the silhouette definition to a record whose mean distance to its own cluster is 4 and to the nearest other cluster is 10.

A) 0.6 <-- correct

B) 0.4

C) 1.5

D) 0.24

Answer: A. The silhouette is (b - a) / max(a, b) = (10 - 4) / 10 = 0.6, indicating a comfortably placed record.

Q22. Compute the purity of a cluster holding 50 records in which the most common true label appears 40 times.

A) 0.4

B) 1.25

C) 0.8 <-- correct

D) 0.2

Answer: C. Purity for a cluster is the share of its records carrying its most frequent label: 40 / 50 = 0.8.

Q23. Use the Adjusted Rand Index to compare two clusterings and determine what a value of 1 signifies.

A) The two partitions share no record at all

B) The two partitions are identical up to renaming <-- correct

C) The two partitions each hold exactly one cluster

D) The index has been computed incorrectly

Answer: B. Cluster names are arbitrary, so the index compares which records are grouped together rather than what the groups are called.

Q24. Solve the following: a segmentation is rebuilt on 10 bootstrap samples and one particular segment reappears in 9 of them. Determine its stability score.

A) 0.1

B) 9.0

C) 0.09

D) 0.9 <-- correct

Answer: D. The score is the proportion of resamples in which the segment recurs: 9 / 10 = 0.9, which is strong evidence that it is genuine.

Q25. Apply an external validation measure and determine what it requires before it can be computed.

A) A second run of the same algorithm

B) A reduction of the data to two dimensions

C) Known labels for at least a subset of records <-- correct

D) An equal number of records in every cluster

Answer: C. External measures score the clustering against a reference grouping, so some ground truth has to exist even if only for a sample.

Q26. Examine a clustering that reports a high silhouette score but whose three clusters hold 995, 3 and 2 records, and determine what to report.

A) The partition is excellent and should be used
B) The silhouette score must be recomputed
C) The number of clusters should be increased
D) The partition is degenerate despite the score <-- correct

Answer: D. Two tiny well-separated outlier groups can lift the average silhouette while the partition says nothing useful about the 995 records that matter.

Q27. Compare internal validation with external validation on what each of the two needs.

A) Internal needs geometry alone, external needs labels <-- correct
B) External needs geometry alone, internal needs labels
C) Both of the two need a set of reference labels
D) Both of the two need only the geometry of the data

Answer: A. This is why internal measures are the everyday tool in unsupervised work, and external ones appear only when a labelled subset happens to exist.

Q28. Use cohesion and separation to determine which of two partitions of the same data is the better.

A) The one with the larger number of clusters
B) The one with tighter clusters that are further apart <-- correct
C) The one with the smaller number of clusters
D) The one produced by the faster algorithm

Answer: B. Lower within-cluster spread and greater between-cluster distance is precisely what the silhouette and the Calinski-Harabasz index both reward.

Q29. Compute the number of record pairs the Rand Index has to consider for a dataset of 10 records.

A) 100
B) 90
C) 10
D) 45 <-- correct

Answer: D. The index examines every unordered pair: 10 x 9 / 2 = 45.

Q30. Apply a stability check to a clustering and determine the correct procedure.

A) Run the algorithm once with a fixed random seed
B) Re-cluster many subsamples and measure the agreement <-- correct
C) Increase k until the inertia stops falling
D) Compare the result against the training labels

Answer: B. Structure that keeps reappearing across resamples is far more likely to be real than structure that changes each time the sample does.

Q31. Use a cluster profile to give a segment a business name.

A) Use the cluster number the algorithm assigned
B) Choose the record nearest the cluster centre
C) Compare its feature averages with the overall averages <-- correct
D) Count how many records the cluster contains

Answer: C. A segment whose visit frequency is high and whose basket size is small becomes 'frequent small-basket shoppers' only once the profile is read this way.

Q32. Examine a segmentation whose groups differ on one feature alone and determine the first thing to check.

A) Whether the features were placed on a common scale <-- correct
B) Whether the algorithm converged successfully
C) Whether the number of records was sufficient
D) Whether the random seed had been recorded

Answer: A. A single wide-ranging unscaled column dominates every distance and produces exactly this symptom. Scaling is the first correction to try.

Q33. Compare Principal Component Analysis with t-SNE for the panels of a visualisation dashboard.

A) PCA gives reproducible axes, t-SNE clearer local groups <-- correct
B) t-SNE gives reproducible axes, PCA clearer local groups
C) Both of the two give axes that can be interpreted
D) Neither of the two is suitable for a dashboard

Answer: A. PCA axes carry explained-variance figures and are stable across runs; t-SNE usually shows neighbourhood structure more vividly but changes with the seed.

Q34. Apply the unsupervised workflow and determine the stage at which the segments are given their business names.

A) Before any of the features are chosen
B) While the distance measure is selected
C) During the standardisation of the columns
D) After the clusters have been profiled <-- correct

Answer: D. Naming requires knowing how each group differs from the rest, which only the profiling step reveals.

Q35. Solve the following: a record has a mean distance of 6 to its own cluster and 6 to the nearest other cluster. Determine its silhouette value.

A) 1
B) 0 <-- correct
C) -1
D) 0.5

Answer: B. The silhouette is (6 - 6) / 6 = 0, meaning the record sits exactly on the boundary between the two clusters.

Q36. Use the Jaccard index to measure how far a segment survives a resample when 40 records are shared and 60 records appear in at least one version.

A) 0.4

B) 1.5

C) 0.667 <-- correct

D) 0.24

Answer: C. The Jaccard index is the shared count divided by the union: 40 / 60 = 0.667, a moderately stable segment.

Q37. Examine an insurer that clusters customers on postcode and behaviour and then offers different premiums by cluster, and determine the principal risk.

A) The clustering will converge too slowly to use

B) The silhouette score will be too low to report

C) Indirect discrimination through a proxy variable <-- correct

D) The number of clusters will be hard to choose

Answer: C. Postcode correlates with protected characteristics, so pricing by cluster can reproduce a prohibited distinction without ever recording the attribute itself.

Q38. Demonstrate how a completed clustering should be handed over to a business team.

A) With the cluster labels and nothing further

B) With profiles, sizes, validation figures and stated limits <-- correct

C) With the full distance matrix of the records

D) With the source code that produced the result

Answer: B. The team needs to know what each segment is, how large it is, how well supported it is, and what the result cannot be used for.

Q39. Relate the interval at which a segmentation is rebuilt to the rate at which the behaviour it describes changes.

A) Faster-changing behaviour calls for fewer rebuilds

B) The interval is fixed once the model is deployed

C) The interval depends only on the number of records

D) Faster-changing behaviour calls for more frequent rebuilds <-- correct

Answer: D. The rebuild schedule should follow the data, and drift monitoring is what tells you when the previous segmentation has stopped describing the population.

Q40. Use two clusterings of the same data to determine what an external measure reports when one has simply had its cluster labels renamed.

A) Perfect agreement, since the groupings are identical <-- correct

B) No agreement, since the labels are all different

C) Partial agreement, in proportion to the renaming

D) The measure cannot be computed in this case

Answer: A. External measures compare which records are placed together, not the arbitrary names given to the groups, so renaming has no effect at all.

----- End of Sample Question Set -----